

Measurement and Comparison of Thermal Conductivity in Metallic Rods Using Infrared Imaging and Digital Thermometry

Group 2E

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1 Introduction

Thermal conductivity (k) is a measure of how fast heat is conducted through a material, typically reported in watts per meter-Kelvin (W/m·K). Materials with higher k transfer heat more quickly, whereas lower k indicates slower heat transfer [1]. Metals are generally good conductors of heat, but there is a wide range among different metals. Copper and aluminum exhibit much higher thermal conductivities (~ 385 and ~ 237 W/m·K, respectively) than stainless steel (~ 16 W/m·K), making them more efficient heat conductors [2].

In this experiment, our group aimed to measure the temperature of three different metal rods using a thermal imager to observe how they heated up over time. The equipment used included two ring stands, a soldering iron, three different metal rods (copper, stainless steel, and aluminum), a Fluke Ti10 thermal imager, and a thermocouple thermometer.

The thermal imager allowed us to save thermal images to a 2 GB SD card. The Fluke Ti10 has a field of view of $23^\circ \times 17^\circ$, which is relatively narrow, but since we were observing a small section of each rod, this specification was suitable for our application. The narrower field also allowed us to take close-up images with improved detail. The Ti10 has a maximum temperature measurement capacity of 482°C , although our experiment did not involve heating the rods to that extent. Nevertheless, knowing the upper limit is helpful for understanding the device's capability [3].



Figure 1: Ti10 Thermal Imager [4]

The image above shows the thermal imager used in the lab. The bluish-green button allowed us to capture thermal images, and by pressing one of the blue side buttons near the display screen, we could save those images to memory. During the experiment, the imager displayed the surface temperature of the rods as they were heated with the soldering iron. We recorded these temperatures at different time intervals to analyze the heating behavior over time.

To verify the accuracy of the thermal imager, we used a thermocouple thermometer as a reference device. Although not shown exactly, the thermometer was similar to the one depicted in the figure below. The thermocouple was placed in direct contact with the rod's surface to measure its temperature in degrees Celsius. For comparison with the infrared camera, we converted these readings to degrees Fahrenheit where necessary.



Figure 2: Thermocouple Thermometer [5]

2 Methods

In this experiment, we have used a Fluke ti-10 infrared camera to compare the heat conductivity of copper, aluminum, and steel rods. To be sure that any variations in temperature conduct were caused by material properties rather than geometry, all three rods had the same size. We also used equipment such as a vernier caliper to measure the diameter of our three rods, a measuring tape to measure their lengths, a stopwatch to record the time, clamps to mount the rods onto a stand, and a soldering iron to heat the rods. The Vernier caliper has an uncertainty of ± 0.01 mm. The stopwatch has a ± 0.5 s uncertainty. The measuring tape has an uncertainty of ± 0.5 mm.



Figure 3: Fluke Ti-10 infrared camera.

First, we mounted clamps with soldering on the stand. Then we attached the rod's one end to the soldering for heating. To get precise surface-temperature measurements, we calibrated the infrared camera, and a thermocouple was securely fastened to the heated end of the rod. The infrared camera was positioned perpendicular to the rods and adjusted so that the rod was clearly visible within the picture before the heating process started. We placed white cardboard behind our setup so that the infrared camera could give us a clear image and consistent background reading. This makes it easier for us to identify the rods on the infrared camera without being distracted by nearby objects. By comparing the camera's readings at the heated end with the thermocouple measurements, the emissivity setting was adjusted. We recorded the initial temperature on both the infrared camera and the thermocouple before starting the experiment. Then we heated the rod for a specific period, and the temperature was continuously recorded by the infrared camera and thermocouple. To guarantee precise timing, we used a stopwatch. We have taken 3 trials and recorded data in our notebook. We repeated the same procedure with our different rods and noted the data.



Figure 4: Fluke Ti-10 infrared camera.

3 Results

The temperature measurements in this experiment were obtained using two instruments: the Fluke Ti-10 infrared (IR) camera and a digital thermometer. The Fluke Ti-10 is a digital infrared imaging device capable of measuring surface temperatures without contact. According to the manufacturer specifications, the temperature measurement accuracy of the IR camera is approximately $\pm 2^{\circ}\text{C}$. The digital thermometer used to verify the IR readings is also a digital device with a measurement precision of $\pm 0.1^{\circ}\text{C}$ and an accuracy of approximately $\pm 1^{\circ}\text{C}$.

The digital thermometer provided the direct-contact temperature of the heated end of each metallic rod, while the IR camera provided a non-contact measurement at the same location. The IR data were recorded in degrees Fahrenheit and later converted to degrees Celsius for consistency with the thermometer measurements.

Tables 1, 2, and 3 summarize the recorded temperature values for copper, aluminum, and stainless steel, respectively.

Table 1: Temperature data for copper rod.

Time (s)	IR Temperature (°F)	IR Temperature (°C)	Digital Thermometer (°C)
0	70.1	21.17	20
15	73.5	23.06	22
30	77.4	25.22	30
45	84.2	29.00	47
60	89.5	31.94	66

Table 2: Temperature data for aluminum rod.

Time (s)	IR Temperature (°F)	IR Temperature (°C)	Digital Thermometer (°C)
0	70.1	21.17	19
15	72.7	22.61	24
30	78.2	25.67	38
45	84.2	29.00	57
60	87.9	31.06	68

Table 3: Temperature data for stainless steel rod.

Time (s)	IR Temperature (°F)	IR Temperature (°C)	Digital Thermometer (°C)
0	70.1	21.17	20
15	70.9	21.61	22
30	73.5	23.06	25
45	76.4	24.67	33
60	75.9	24.39	42

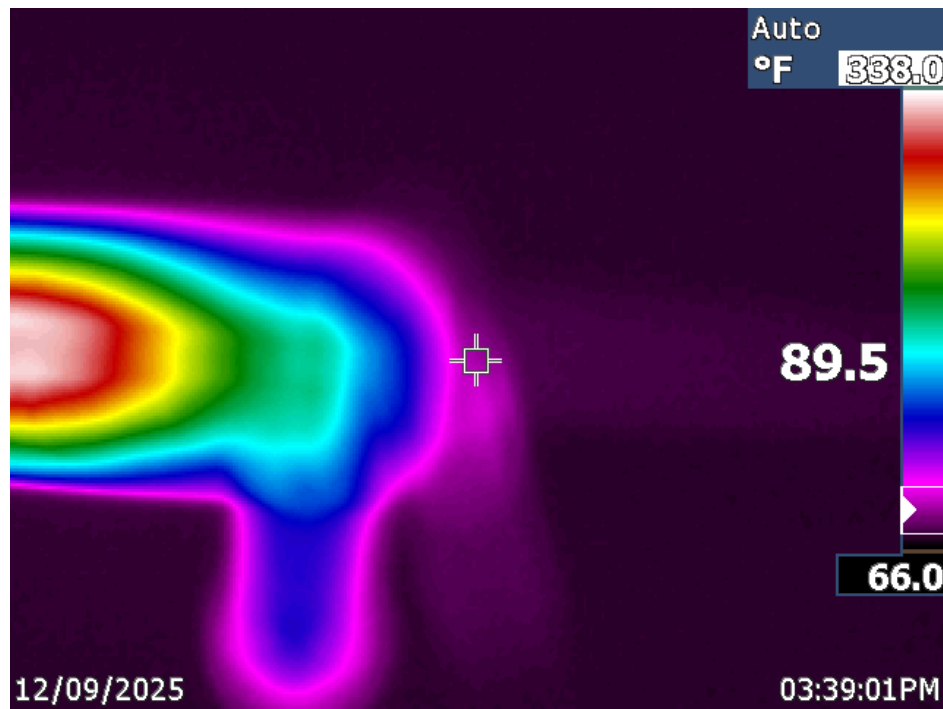


Figure 5: Picture captured from an infrared camera showing temperature of a copper rod after 60 seconds of heating

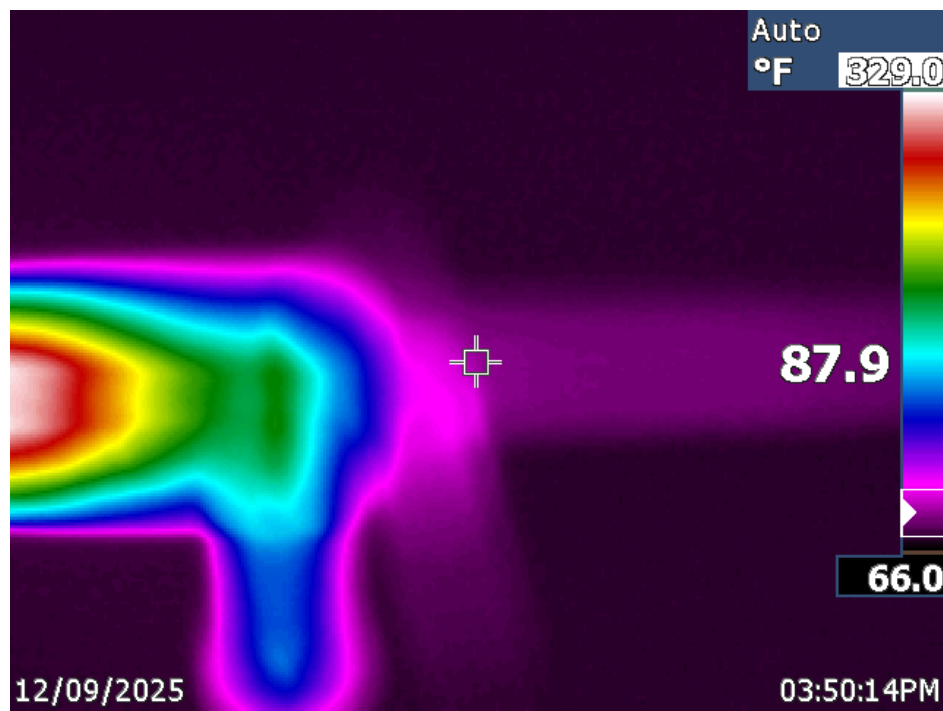


Figure 6: Picture captured from an infrared camera showing temperature of an aluminum rod after 60 seconds of heating

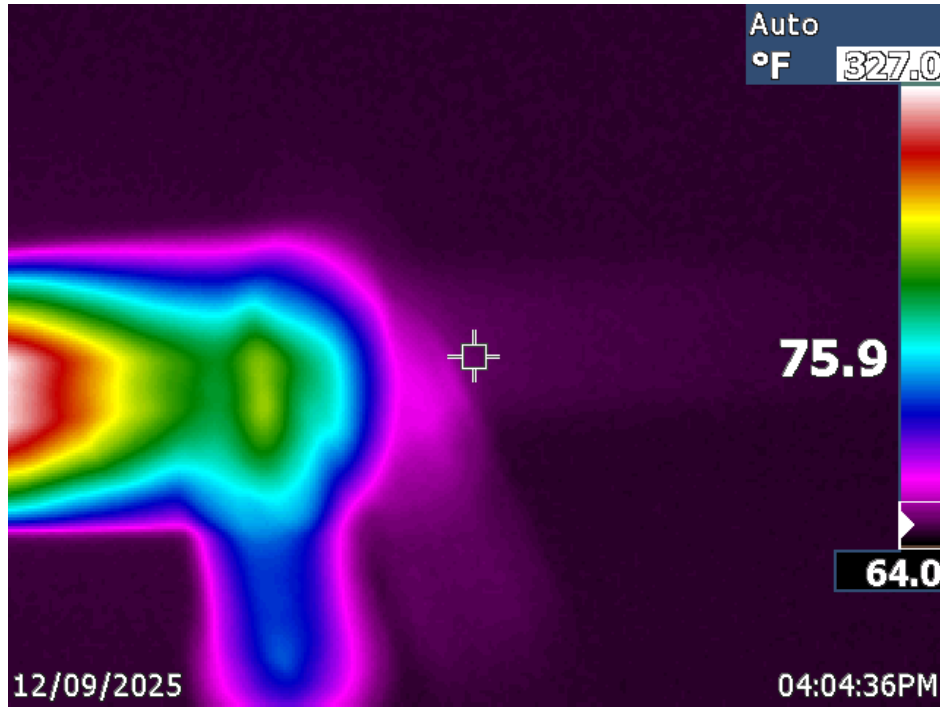


Figure 7: Picture captured from an infrared camera showing temperature of a stainless steel after 60 seconds of heating

Across all trials, the digital thermometer measured substantially higher temperatures than the IR camera, particularly for copper and aluminum. This discrepancy can be attributed primarily to the low emissivity of polished metallic surfaces, which causes the IR camera to underestimate temperature even after emissivity calibration. The effect is most pronounced in copper and aluminum, both of which have highly reflective surfaces. Stainless steel, which has a higher emissivity than the other two metals, exhibited smaller deviations between IR and digital readings.

In addition to emissivity-related errors, the rods used in this experiment had very small diameters (approximately 3 mm). Such narrow targets occupy only a few pixels in the infrared image, making it difficult for the camera to accurately isolate and measure the rod's temperature. The limited spatial resolution of the IR camera further contributed to measurement inaccuracy, especially at higher temperatures when thermal gradients were steep.

4 Analysis and Discussion

All quantitative calculations, including linear regression, heating rate determination, infrared error evaluation, and normalized heating-rate comparisons, were performed in a Python environment. A separate Jupyter Notebook containing the complete code and output has been submitted alongside this report. In this section, we focus on interpreting the behavior shown in the plotted results and relating the observed trends to the physical properties of the materials.

To quantify the rate at which each material heated during the experiment, a linear regression was applied to the digital thermometer temperature data using the model

$$T(t) = mt + b,$$

where m represents the heating rate in $^{\circ}\text{C/s}$. The regression was performed in Python using the `LinearRegression` function from the `scikit-learn` library. The resulting heating rates were:

Copper: $0.780\text{ }^{\circ}\text{C/s}$, Aluminum: $0.873\text{ }^{\circ}\text{C/s}$, Stainless Steel: $0.367\text{ }^{\circ}\text{C/s}$.

These values align with the expected thermal behavior of the materials. Aluminum and copper exhibited significantly larger heating rates than stainless steel, reflecting their higher thermal conductivities. Although aluminum showed a slightly higher heating rate than copper in the raw data, this discrepancy is likely due to geometric differences (e.g., varying rod diameters), small timing uncertainties,. Stainless steel's much lower heating rate confirms its well-known poor thermal conductivity relative to copper and aluminum.

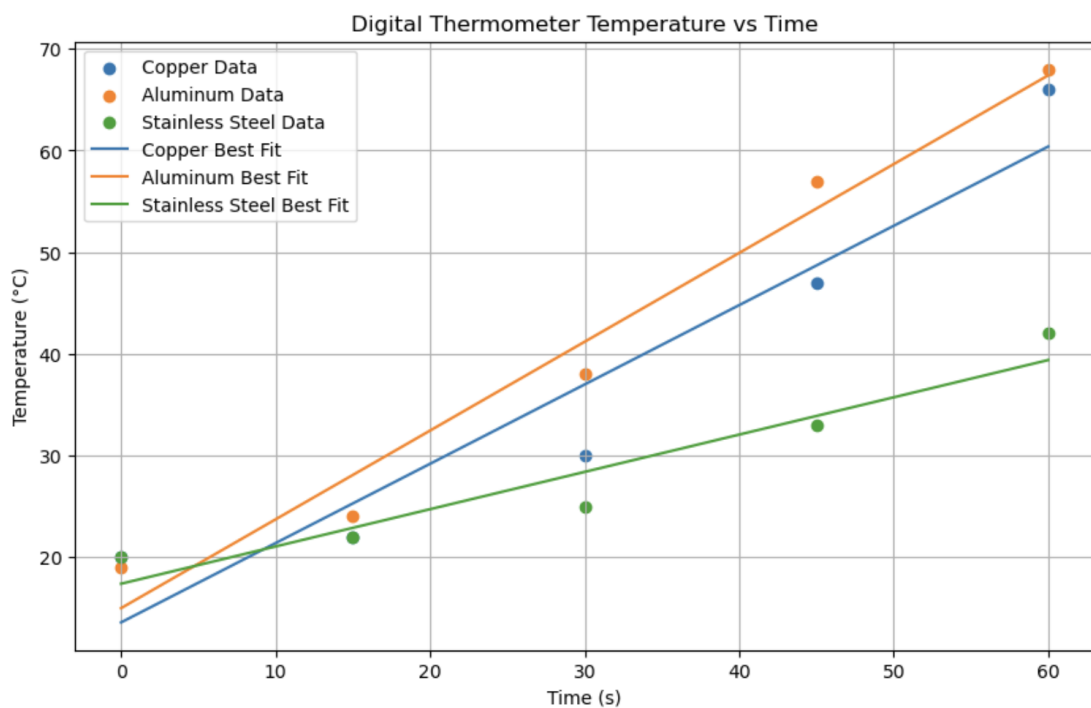


Figure 8: Digital thermometer temperature–time graph.

The digital thermometer temperature–time graphs demonstrated a clear linear increase in temperature for all three materials during the heating period. Copper and aluminum showed the steepest slopes, indicating rapid heat transfer along the rod. Stainless steel exhibited a much shallower slope, confirming its significantly lower thermal conductivity. The linear regression fits overlaid on the scatter data revealed strong agreement, validating the use of a linear model over the short 60 s heating interval.

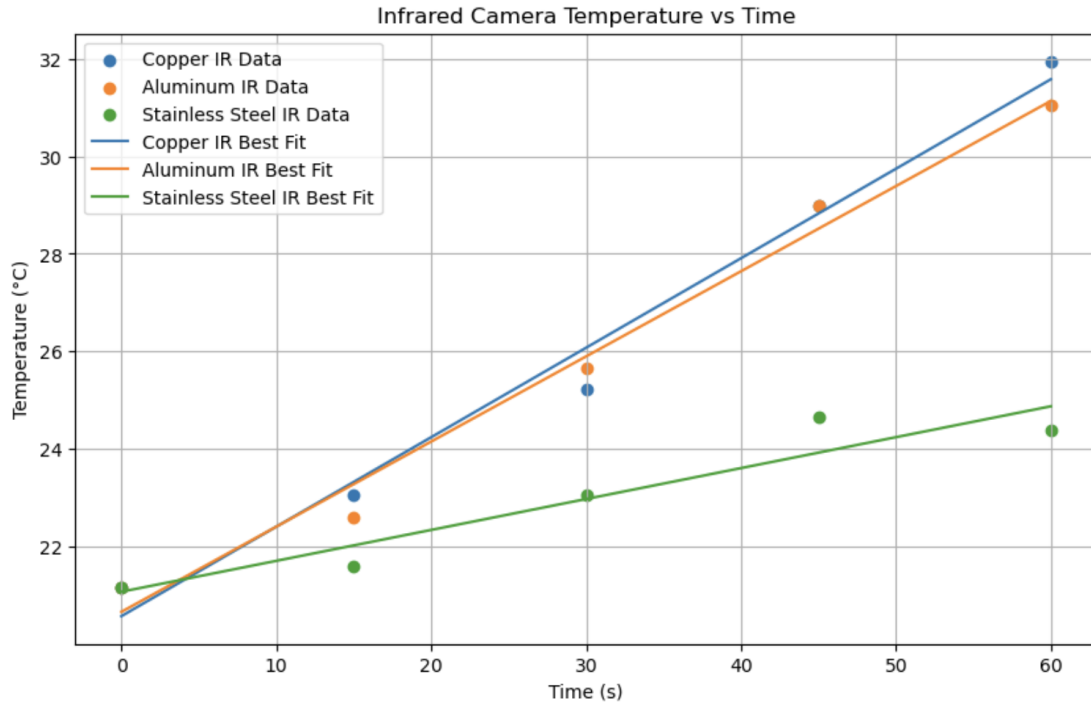


Figure 9: Infrared camera temperature–time graph.

When the infrared temperature data were plotted, the overall trends remained consistent across materials, but the absolute temperature values were significantly lower than those measured by the digital thermometer. The regression lines for the IR data displayed noticeably reduced slopes compared to the thermometer data. This reflects the IR camera’s difficulty in accurately measuring low-emissivity metallic surfaces and the additional spatial-resolution limitations caused by the very small rod diameter.

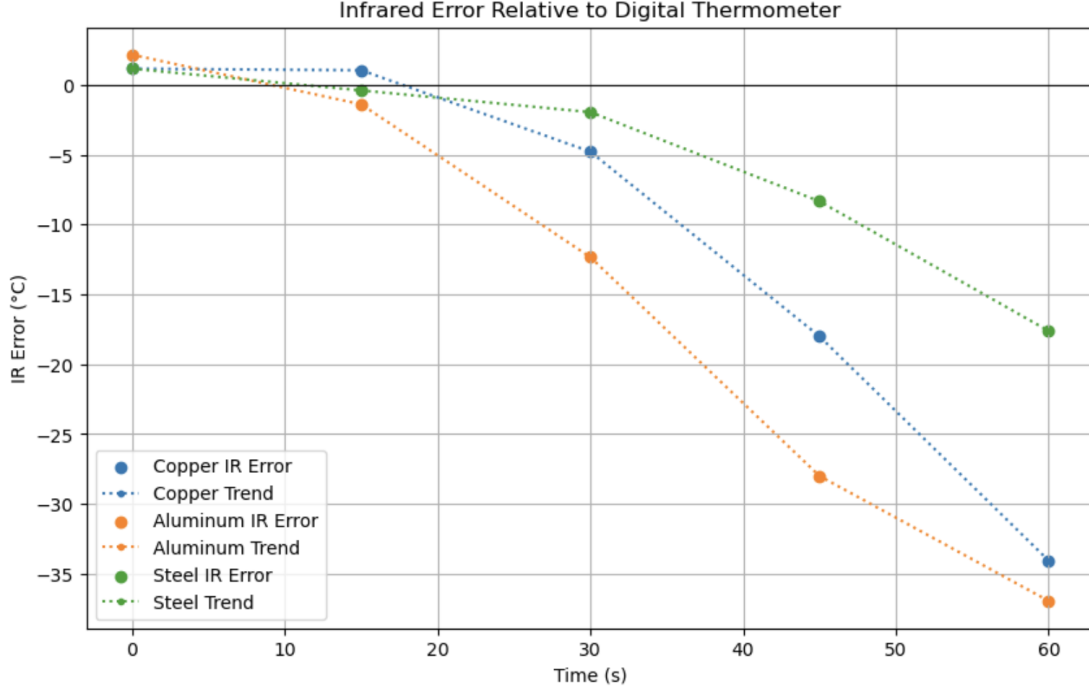


Figure 10: Infrared camera temperature–time graph.

The IR error plot provided further insight into the performance of the infrared camera. The error was computed at each time point using

$$\text{Error} = T_{\text{IR}} - T_{\text{digital}},$$

where T_{IR} is the infrared temperature (converted from Fahrenheit to Celsius), and T_{digital} is the direct-contact thermometer reading. A negative error value therefore indicates that the IR camera reported a temperature lower than the actual surface temperature measured by the digital thermometer.

Across all three materials, the error became increasingly negative as time progressed, demonstrating that the infrared camera consistently underestimated the rod temperature, with the magnitude of the underestimation growing at higher temperatures. The average IR errors calculated from the Python analysis were:

$$\text{Copper: } -10.92^{\circ}\text{C}, \quad \text{Aluminum: } -15.30^{\circ}\text{C}, \quad \text{Stainless Steel: } -5.42^{\circ}\text{C}.$$

Aluminum exhibited the largest deviation, followed by copper, while stainless steel showed the smallest. These trends correlate directly with material emissivity: polished aluminum and copper surfaces emit less thermal radiation and reflect more ambient infrared energy, leading to substantial measurement underestimation. Stainless steel, with comparatively higher emissivity, produced smaller errors. The small rod diameter further contributed to IR inaccuracy, as the rod occupied only a limited number of pixels in the thermal image, restricting the camera's ability to resolve its true temperature.

To better compare the heating behavior of the three materials, the heating rates obtained from the linear regression were normalized by the cross-sectional area of each rod. This normalization accounts for geometric differences, particularly the larger diameter of the copper rod, which increases its thermal mass and reduces its apparent heating rate. The cross-sectional area of each rod was computed using

$$A = \pi \left(\frac{d}{2} \right)^2,$$

where d is the measured diameter of the rod. The normalized heating rate was then calculated as

$$\frac{m}{A},$$

where m is the heating rate in $^{\circ}\text{C/s}$ from the regression model. This quantity represents the temperature rise per unit area and provides a more fundamental comparison of how efficiently each material conducts heat independent of rod size.

Using the measured diameters, the areas were found to be $8.76 \times 10^{-6} \text{ m}^2$ for copper and $6.83 \times 10^{-6} \text{ m}^2$ for both aluminum and stainless steel. Combining these areas with the experimentally determined heating rates yielded the following normalized heating values:

Copper: $8.90 \times 10^4 \text{ }^{\circ}\text{C s}^{-1} \text{ m}^{-2}$, Aluminum: $1.28 \times 10^5 \text{ }^{\circ}\text{C s}^{-1} \text{ m}^{-2}$, Stainless Steel: $5.36 \times 10^4 \text{ }^{\circ}\text{C s}^{-1} \text{ m}^{-2}$.

These normalized values emphasize the expected trend: both copper and aluminum conduct heat far more effectively than stainless steel. Although aluminum still appears to outperform copper in the normalized comparison, this is not physically meaningful and is

likely due to experimental uncertainties such as imperfect thermal contact, small timing deviations, and the short heating duration of the experiment. The normalization nonetheless helps mitigate geometric effects and provides a clearer comparison of the materials' intrinsic thermal behavior.

5 Conclusion:

This experiment successfully demonstrated the thermal conductivity differences among copper, aluminum, and stainless steel rods using both infrared thermography and a digital thermometer. The temperature–time profiles obtained through the digital thermometer confirmed expected material behavior: copper and aluminum heated rapidly due to their high thermal conductivities, while stainless steel exhibited a significantly slower rate of temperature increase. However, the infrared camera consistently underestimated surface temperatures, particularly for copper and aluminum. This discrepancy is attributed to the low emissivity and limited resolution of the IR camera when measuring narrow cylindrical targets. Despite emissivity calibration, the IR measurements remained less accurate than contact-based readings, especially at higher temperatures.

Overall, the experiment emphasized the importance of material properties, surface characteristics, and measurement methodology when evaluating thermal performance. It also highlighted the practical limitations of infrared thermography in accurately assessing temperature on small, low-emissivity metallic objects. Future studies could improve accuracy by applying high-emissivity coatings or using IR cameras with higher spatial resolution.

References

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